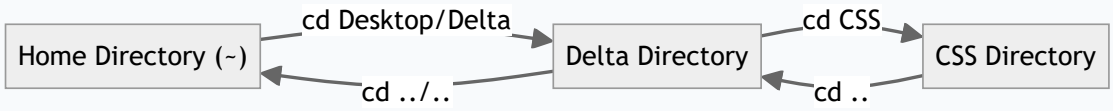
Submodule 34.06: Directory Navigation: `cd`, `..`, Chaining Paths, and Autocomplete

Topic Overview & Directory Traversal

In a graphical interface, navigating into a folder requires double-clicking its icon, while navigating backward requires clicking the "Back" arrow. In the terminal, traversal is executed using the **`cd` (Change Directory)** command.



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1. Core Navigation Commands & Syntax

A. Stepping Inward: `cd <directory_name>`

To navigate into a child directory located within your current working directory:

```
# Move into Desktop
cd Desktop

# Move into Delta folder
cd Delta
```

B. Stepping Backward: `cd ..`

The double dot `..` represents the **parent directory** (one level up in the file hierarchy).

```
# Go back one level up (equivalent to GUI back button)
cd ..
```

C. Deep Traversal (Path Chaining)

Instead of executing sequential single-level `cd` commands, paths can be chained using forward slashes (`/`):

```
# Move forward multiple levels in a single command
cd Desktop/Delta/CSS

# Move backward multiple levels in a single command
cd ../..
```

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2. The Tab Autocomplete Feature

Developers rarely type out entire directory names.

- Type the first 2–3 characters of a folder name and press the **Tab** key.
- The shell automatically completes the full directory or file name.
- If multiple candidates exist, tapping Tab twice displays all matching options.
- *Best Practice*: Always use Tab autocomplete to prevent typographical errors.

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3. Error Handling: Non-Existent Directories

If you attempt to navigate into a folder that does not exist or mistype its name:

```
cd NonExistentFolder
# bash: cd: NonExistentFolder: No such file or directory
```

Resolution: Run `ls` to verify the exact case-sensitive spelling of target directories before executing `cd`.